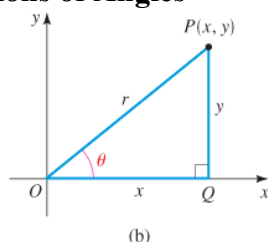
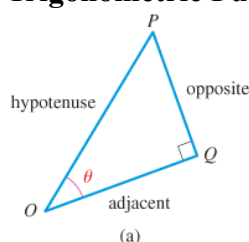


Chp 5.3: Trigonometric Functions of Angles**I. Trigonometric Functions of Angles****Definition of the Trigonometric Functions**

Let θ be an angle in standard position and let $P(x, y)$ be a point on the terminal side. If

$r = \sqrt{x^2 + y^2}$ is the distance from the origin to the point $P(x, y)$, then

$$\sin \theta = \frac{y}{r} = \frac{O}{H} = \frac{1}{\csc \theta}$$

$$\cos \theta = \frac{x}{r} = \frac{A}{H} = \frac{1}{\sec \theta}$$

$$\tan \theta = \frac{y}{x} \quad (x \neq 0) = \frac{O}{A} = \frac{\sin \theta}{\cos \theta} = \frac{1}{\cot \theta}$$

$$\csc \theta = \frac{r}{y} \quad (y \neq 0)$$

$$\sec \theta = \frac{r}{x} \quad (x \neq 0) = \frac{H}{A} = \frac{1}{\cos \theta}$$

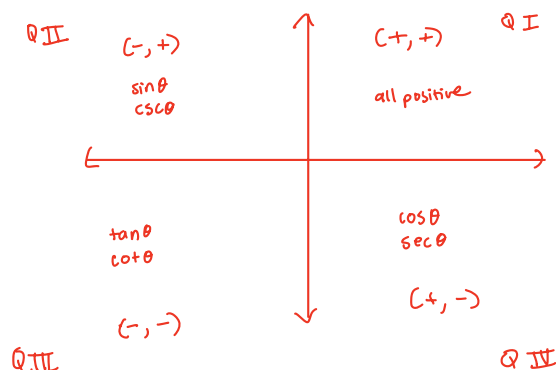
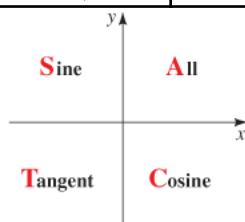
$$\cot \theta = \frac{x}{y} \quad (y \neq 0) = \frac{A}{O} = \frac{\cos \theta}{\sin \theta} = \frac{1}{\tan \theta}$$

The angles for which the trigonometric functions may be undefined are the angles for which either the x - or y -coordinate of a point on the terminal side of the angle is 0. These angles are called quadrantal angles, that is, angles that are coterminal with the coordinate axes. It is a crucial fact that the values of the trigonometric functions do not depend on the choice of the point $P(x, y)$.

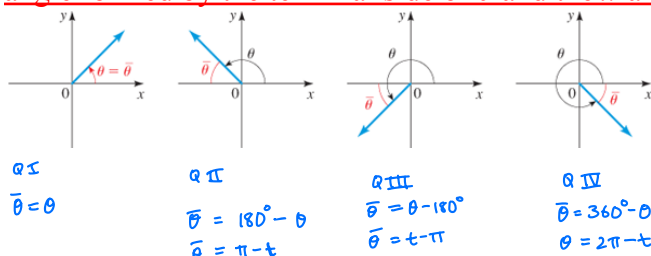
II. Evaluating Trigonometric Functions at Any Angle**Signs of the Trigonometric Functions**

Complete the following table to indicate which trigonometric functions are positive and which are negative in each quadrant.

Quadrant	Positive Functions	Negative Functions
I	<u>all</u>	<u>none</u>
II	<u>sin, csc</u>	<u>cos, sec, tan, cot</u>
III	<u>tan, cot</u>	<u>sin, csc, cos, sec</u>
IV	<u>cos, sec</u>	<u>sin, csc, tan, cot</u>

**Reference Angle**

Let θ be an angle in standard position. The reference angle $\bar{\theta}$ associated with θ is the acute angle formed by the terminal side of θ and the x -axis.



Evaluating Trigonometric Functions for Any Angle

To find the values of the trigonometric functions for any angle θ , we carry out the following steps

1. Find the reference angle $\bar{\theta}$ associated with the angle θ .
2. Determine the sign of the trigonometric function of θ by noting the quadrant in which θ lies.
3. The value of the trigonometric function of θ is the same, except possibly for sign, as the value of the trigonometric function of $\bar{\theta}$.

III. Trigonometric Identities

The trigonometric functions of angles are related to each other through several important equations called trigonometric identities.

Fundamental Identities

Reciprocal Identities

$$\csc \theta = \frac{1}{\sin \theta}, \sec \theta = \frac{1}{\cos \theta}, \cot \theta = \frac{1}{\tan \theta}, \tan \theta = \frac{\sin \theta}{\cos \theta}, \cot \theta = \frac{\cos \theta}{\sin \theta}$$

Pythagorean Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\cos^2 \theta = 1 - \sin^2 \theta$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$\tan^2 \theta = \sec^2 \theta - 1$$

$$1 = \sec^2 \theta - \tan^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$1 = \csc^2 \theta - \cot^2 \theta$$

$$\cot^2 \theta = \csc^2 \theta - 1$$

IV. Areas of Triangles

The area A of a triangle with sides of lengths a and b and with included angle θ is

$$A = \frac{1}{2} ab \sin \theta$$

Example 1: Find the reference angle for the given angle.

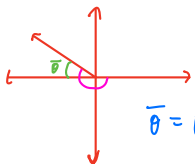
a. -199°

b. $\frac{10\pi}{9}$

c. -1.4π

d. 2.3

e. 485°



$$\bar{\theta} = 199^\circ - 180^\circ = 19^\circ$$

$$\bar{\theta} = \frac{\pi}{9}$$

$$\bar{\theta} = 1.4\pi - \pi = 0.4\pi$$

$$\bar{\theta} = \pi - 2.3$$

Example 2: Find the exact value of the trigonometric function.

a. $\csc 135^\circ$

b. $\sec -420^\circ$

c. $\tan -\frac{4\pi}{3}$

d. $\cos \frac{7\pi}{4}$

e. $\sin \frac{11\pi}{6}$

$$\csc 135^\circ = \frac{1}{\sin 135^\circ} = \sqrt{2}$$

$$\sec -420^\circ = \frac{1}{\cos -420^\circ} = 2$$

$$\tan -\frac{4\pi}{3} = -\sqrt{3}$$

$$\cos \frac{7\pi}{4} = \frac{\sqrt{2}}{2}$$

Example 3: From the information given, find the quadrant that θ lies in.

a. $\tan \theta < 0$ and $\sin \theta < 0$

b. $\csc \theta > 0$ and $\cos \theta < 0$

$$\tan \theta, - \quad \sin \theta, -$$

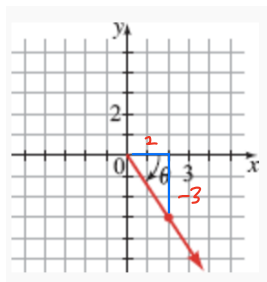
Q IV ← both negative

$$\boxed{\text{Q II}}$$

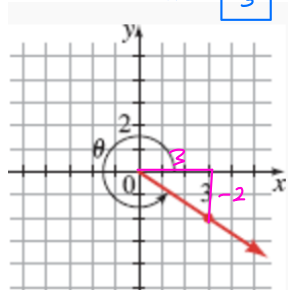
All students take calculus

Example 4: An angle θ is given in the coordinate plane. Find the exact value of the indicated trigonometric function of θ . (You may find it helpful to use the Pythagorean Theorem.)

a. $\cot \theta = \frac{A}{O} = \frac{-2}{3}$



b. $\sec \theta = \frac{H}{A} = \frac{\sqrt{13}}{3}$

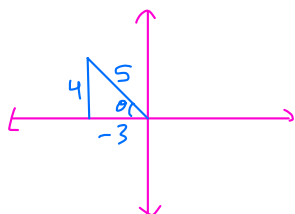


$$r = \sqrt{9 + 4} = \sqrt{13}$$

Example 5: Find the values of the trigonometric functions of θ from the information given.

a. $\cos \theta = \frac{-3}{5}$, θ in Quadrant II

b. $\cot \theta = -2$, θ in Quadrant IV



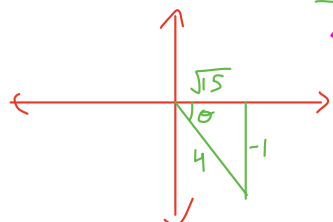
$$\sec \theta = -\frac{5}{3}$$

$$\sin \theta = \frac{4}{5}, \csc \theta = \frac{5}{4}$$

$$\tan \theta = -\frac{4}{3}, \cot \theta = -\frac{3}{4}$$

c. $\csc \theta = -4$, $\tan \theta < 0 \rightarrow Q IV$

d. $\sin \theta = \frac{3}{10}$, $\sec \theta > 0$



$$\sin \theta = \frac{-4}{5}$$

$$\cot \theta = -\sqrt{5}$$

$$\cos \theta = \frac{\sqrt{5}}{5}$$

$$\sec \theta = \frac{5\sqrt{5}}{4}$$

$$\tan \theta = \frac{-\sqrt{5}}{4}$$

Example 6: Write the first trigonometric function in terms of the second for in the given quadrant.

a. $\sec \theta$, $\tan \theta$, Quadrant II

b. $\csc \theta$, $\cot \theta$, Quadrant III

$$\sec^2 \theta = 1 + \tan^2 \theta$$

$$\sec \theta = \pm \sqrt{1 + \tan^2 \theta}$$

$$\sec \theta = -\sqrt{1 + \tan^2 \theta}$$

Example 7: If $\theta = \frac{\pi}{3}$ find the value of each expression.

$\sin^2 \theta$, $\sin(\theta^2)$

$$\begin{aligned} \sin^2 \theta &= (\sin \theta)^2 \\ &= \left(\sin \frac{\pi}{3}\right)^2 \\ &= \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4} \end{aligned}$$

$$\begin{aligned} \sin(\theta^2) &= \sin\left(\left(\frac{\pi}{3}\right)^2\right) \\ &= \sin\left(\frac{\pi^2}{9}\right) \\ &= 0.8897 \end{aligned}$$

Example 8:

- a. A triangle with sides of length 10 and 22 and included angle 10° . Find the area of the triangle.

$$A = \frac{1}{2} (10)(22) \sin 10^\circ$$

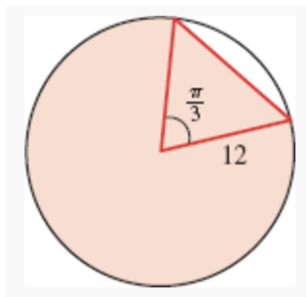
$$A = 19.10 \text{ unit}^2$$

- b. A triangle has area 16 in^2 . The angle between two sides of the triangle is 36° , where one of these two sides has length 5 in. Find the length of the other side.

$$16 = \frac{1}{2} 5 \cdot b \sin 36^\circ$$

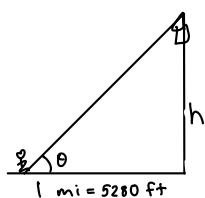
$$\frac{32}{5 \sin 36^\circ} = b \quad \therefore b = 10.89 \text{ in}$$

- c. Find the area of the shaded region in the figure.



Example 9: A rocket fired straight upward is tracked by an observer on the ground, one mile away.

- (a) Show that when the angle of elevation is θ , the height of the rocket (in feet) is $h = 5280 \tan \theta$



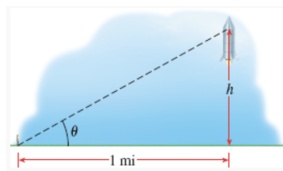
$$\tan \theta = \frac{h}{1 \text{ mi}}$$

$$\tan \theta = \frac{h}{5280 \text{ ft}}$$

$$\therefore h = 5280 \tan \theta$$

- (b) Complete the table to find the height of the rocket at the given angles of elevation.

θ	20°	60°	80°	85°
h	$\approx 1922 \text{ ft}$	$\approx 9145 \text{ ft}$	$\approx 29944 \text{ ft}$	$\approx 60351 \text{ ft}$



$$\theta = 20^\circ:$$

$$h = 5280 \tan(20^\circ) \approx 1922 \text{ ft}$$

$$\theta = 60^\circ:$$

$$h = 5280 \tan(60^\circ) \approx 9145 \text{ ft}$$

$$\theta = 80^\circ:$$

$$h = 5280 \tan(80^\circ) \approx 29944 \text{ ft}$$

$$\theta = 85^\circ$$

$$h = 5280 \tan(85^\circ) \approx 60351 \text{ ft}$$